



Built on the lines of payphones, here is an automatic coin collection device for payloads like lamps and air-conditioners to be used on a private electrical line. It is useful for paying guest houses, lodges and trains.

The system makes use of a sensor for detecting the coin and a micro-controller that counts the coins and shows the count on a 7-segment display. When you close the load switch provided in the circuit, the relay energises to connect the load and the coin count on display starts decrementing. When the count decrements to zero, the relay de-energises to disconnect the load.

Circuit description

Fig. 1 shows the put-coin-and-draw-power circuit. It comprises microcontroller AT89C2051 (IC2), dual timer NE555 (IC1), 7-segment decoder CD4511 (IC3), regulators 7805 and 7806 (IC4 and IC5), and a few discrete components.

LED1 is used as the light source for light-dependent resistor LDR1, which is made of cadmium sulphide and acts as the coin detector. Resistor R1 limits the current through LED1. The light from LED1 falls continuously on LDR1, whose resistance decreases with increase in the incident light intensity.

The NE556 dual monolithic timing circuit is a highly stable controller capable of producing accurate time delays. It is basically a dual NE555. In the time delay mode of operation, the time

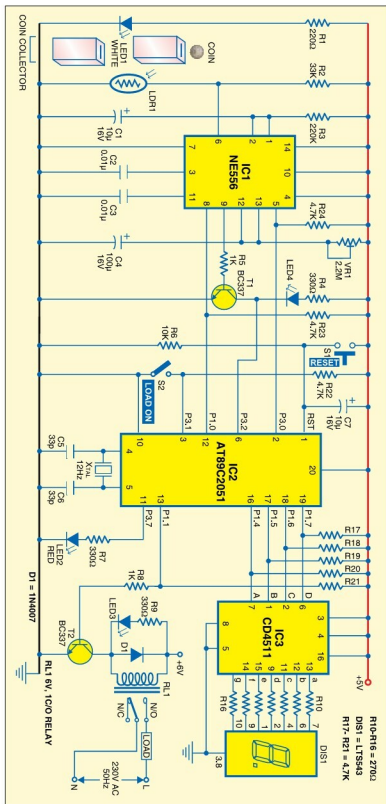


Fig. 1: Put-coin-and-draw-power circuit